

Rafael Figueroa Gomez

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PROFESSIONAL SUMMARY

Electrical and Computer Engineering student at the University of Washington with hands-on experience developing embedded systems, analog front-end circuits, and hardware prototypes for aerospace and medical device applications. Direct experience designing, coding, testing, and troubleshooting embedded C/C++ firmware on microcontroller platforms, with working knowledge of MATLAB/Simulink, Python, and Git. Bilingual in English and Spanish with a strong track record of technical execution, cross-functional collaboration, and academic excellence.

EDUCATION

University of Washington | Seattle, WA

Bachelor of Science, Electrical and Computer Engineering — Expected June 2028

Relevant Coursework: Digital Circuits and Systems, Programming for Signal and Information Processing Applications, Computer Hardware Skills

Whatcom Community College | Bellingham, WA

Associate in Science Transfer Degree, Electrical and Computer Engineering — GPA: 3.95 — June 2026

Relevant Coursework: Circuits, Differential Equations, Linear Algebra, Electromagnetic Physics, Java (OOP), Engineering Numerical Methods

EXPERIENCE

SARP — Avionics/DAQ, University of Washington | Fall 2026 – Present

- Selected to join the Avionics team's DAQ subsystem, contributing to design and bring-up of sensor data acquisition hardware (pressure transducers, thermocouples, accelerometers, IMUs) for flight and static motor fire telemetry, applying embedded systems, analog front-end, and PCB layout skills to hardware/software integration work

Peer Tutor & Peer Mentor, Whatcom Community College | 2025 – 2026

- Guided 15+ peers each quarter through Calculus, Physics, Statics, and Differential Equations, breaking down unfamiliar problems in real time and reducing student anxiety around difficult coursework
- Organized and led engineering outreach events to encourage prospective students to pursue engineering

Coa Mexican Eatery | 2022 – 2025

- Trained and cross-trained staff across 3 roles, reducing onboarding time by 2 weeks
- Led a team of 3 to deliver exceptional customer service, increasing sales by 5%

PROJECTS

Heartbeat Monitor — Analog Front-End, PCB & Embedded Firmware

- Designed a three-stage analog front end (high-pass with gain, low-pass, band-pass with gain) using OP484 op-amps to condition and amplify weak bio signals, with the final band-pass stage isolating the heartbeat frequency range from surrounding noise
- Implemented negative-rail generation (ICL7660S charge pump) for single-supply operation from 5V; verified circuit operation and signal integrity using an EspoTek Labrador portable oscilloscope
- Developing bare-metal firmware in C on a Raspberry Pi Pico for ADC sampling and serial data logging, replacing an earlier Arduino-based acquisition approach
- Transitioning design from breadboard to a custom PCB in Altium Designer for a production-representative single-channel DAQ front end

Negative Pressure Wound Therapy (NPWT) Device

- Debugged, rewrote, and tested undocumented pressure-sensor synchronization firmware from scratch after inheriting a non-functional codebase, using reverse-engineered reference implementations
- Built an operational device for under-resourced clinical settings, integrating Python tooling, analog circuit analysis, and microcontroller firmware; validated through continuous multi-hour operation; encouraged by faculty to present at a state-wide conference

Solar Panel Energy Optimization (MATLAB/Simulink)

- Simulated multiple system parameters in MATLAB/Simulink to evaluate algorithmic approaches for solar panel maximum power point tracking (MPPT), researching dynamic algorithm switching based on operating conditions
- Authored an IEEE-formatted research paper documenting methodology and findings

SKILLS

Embedded & Hardware: Embedded C/C++, Microcontrollers (Arduino, Raspberry Pi Pico — bare-metal), I2C, SPI, UART, Analog Circuit Design, PCB Layout (Altium Designer, KiCad), Oscilloscope/Multimeter, Soldering, 3D Modeling/Printing (Onshape)

Programming & Tools: Python, C++, Java, MATLAB/Simulink, Git (Version Control / Software Configuration Management)

Languages: English (Native), Spanish (Fluent)